

BOOKLET 3 • ELECTRONICS AND ANALOGUE
CONTROL

The Unknown Thermistor

A box of components lost its labels.
Find out what they are, and prove it.

National 5 Engineering Science



Daily Review

Lesson 4 — Input and Output Devices

- 1 What does an input transducer do?
- 2 What does an LDR react to?
- 3 Describe how an NTC thermistor's resistance changes as it gets hotter.
- 4 Name an output transducer that produces sound.
- 5 State the unit of resistance.

PART 1

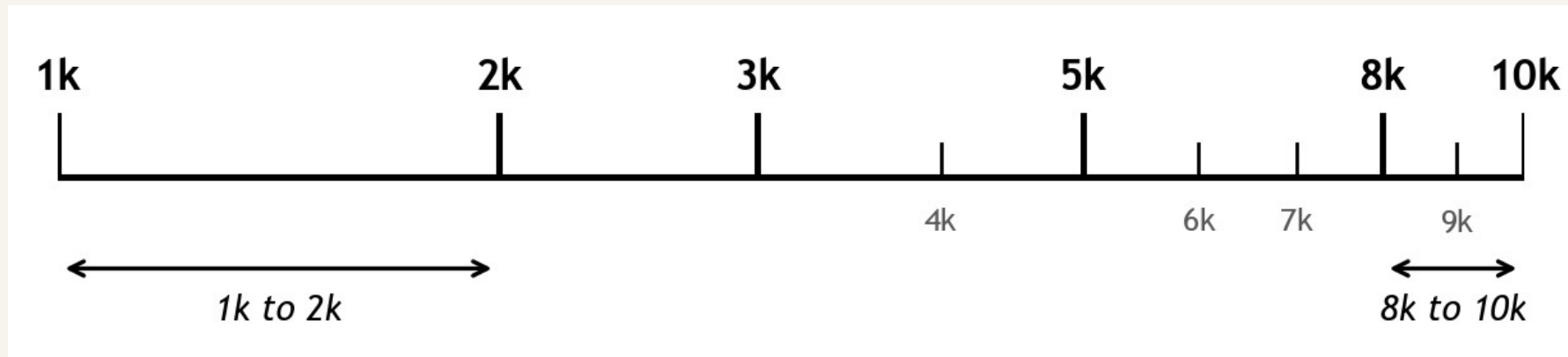
Reading a Characteristic Graph

Before you can say what a component is, you have to be able to read what the maker says it is.

By the end of this part...

- 1 I can explain why a resistance axis is not evenly spaced
- 2 I can read a resistance off a thermistor characteristic graph
- 3 I can work backwards from a resistance to the component

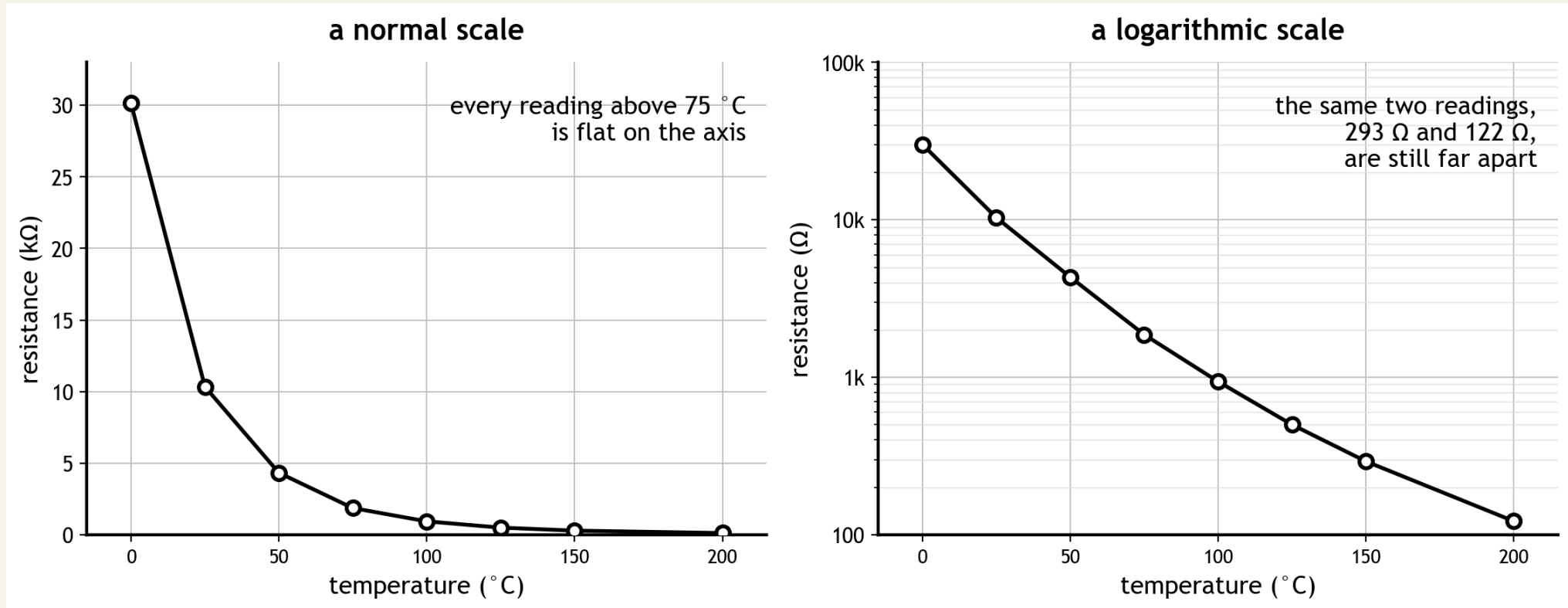
The scale that is not even



One step of a resistance axis, from 1 k Ω to 10 k Ω .

Half the width is used up getting from 1 k Ω to 3 k Ω .
The gap from 8 k Ω to 10 k Ω is tiny.

Why not just use a normal scale?



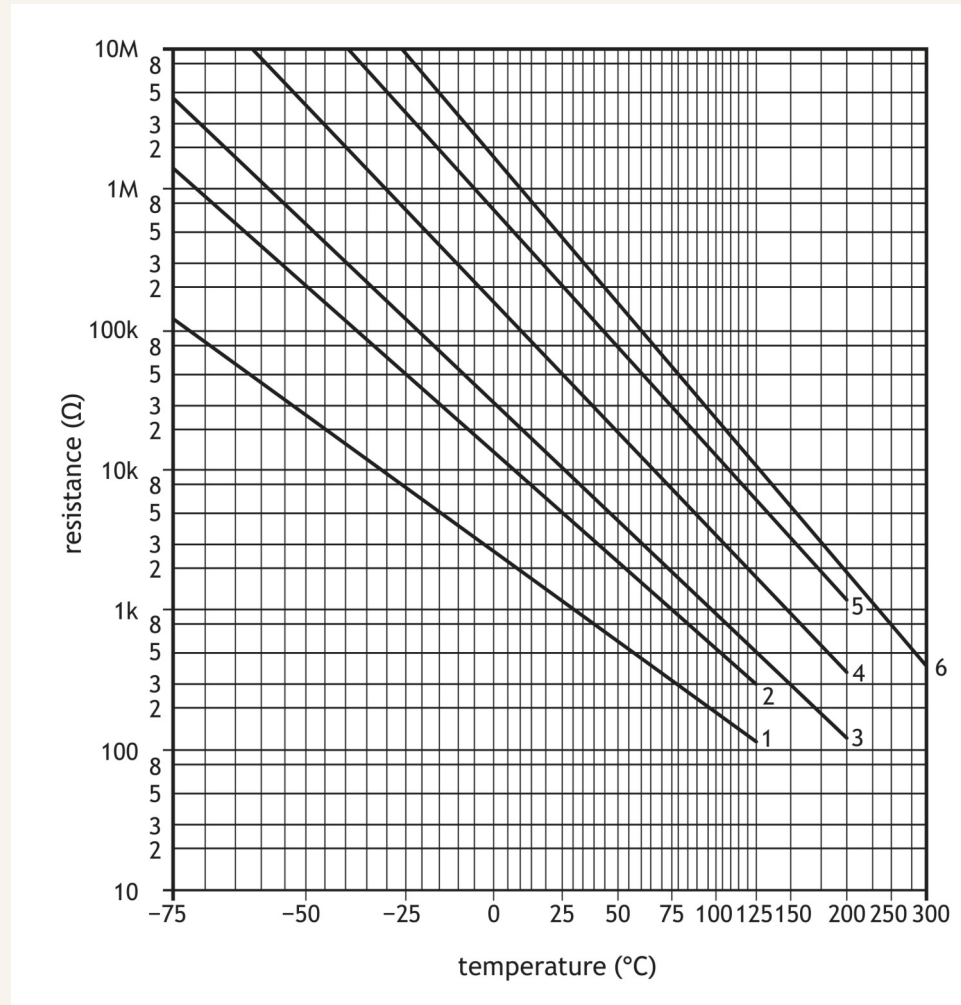
One thermistor, the same eight readings, plotted twice.

The four steps

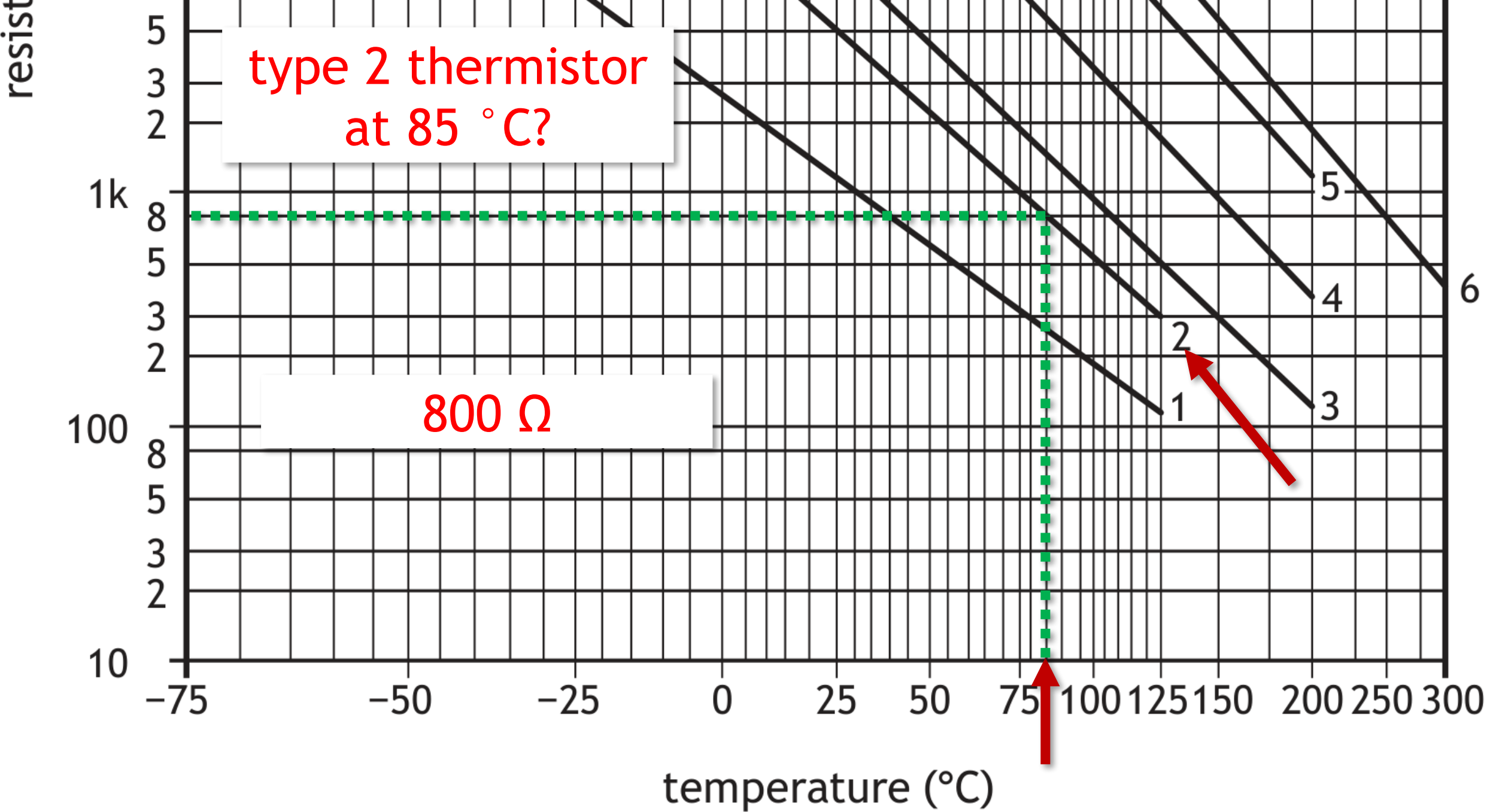
Step	What you do
1	Find the temperature on the bottom axis
2	Rule straight up to the numbered curve
3	Rule straight across to the resistance axis
4	Read the value – and write the unit

Use a ruler – every year candidates lose this by eye alone.

The graph the exam gives you – six thermistors, 1 to 6



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Show me — read it off the graph

SHOW ME BOARDS

- 1 Type 3 at 10 °C.
- 2 Type 1 at 30 °C.
- 3 Type 5 at 50 °C.
- 4 Type 4 at 0 °C.
- 5 Which type is 1 k Ω at 30 °C?

Show me — read it off the graph

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Type 6 at 100 °C.
- 2 Type 2 at 25 °C.
- 3 At what temperature is type 2 equal to 10 kΩ?
- 4 Which type is about 100 kΩ at 25 °C?
- 5 Which type has the highest resistance at any temperature?

Your turn

Sensor Investigation sheet – Stage 1

Work through Stage 1, questions 1 to 12.
Rule every line in pencil. Every answer carries
a unit.

Sensor Investigation sheet, Stage 1

Checkpoint

- 1 Why is a resistance axis not evenly spaced?
- 2 A point sits halfway between 10k and 100k. Why is “55 k Ω ” wrong?
- 3 A supplier says only “10 k Ω ”. What else do you need to know?

Daily Review

Part 1 — Reading a characteristic graph

- 1 State the four steps for reading a value off a characteristic graph.
- 2 What does the unit have to do with the mark?
- 3 Does an NTC thermistor's resistance rise or fall as it warms?
- 4 What is the unit of resistance?
- 5 Why must a reading be taken twice?

PART 2

Planning the Test

Nobody is going to hand you a method. Writing it is the job.

By the end of this part...

- 1** I can choose a suitable range on a manual multimeter
- 2** I can write a method another pupil could follow
- 3** I can identify the hazards in a hot-water test

The brief

THE BRIEF

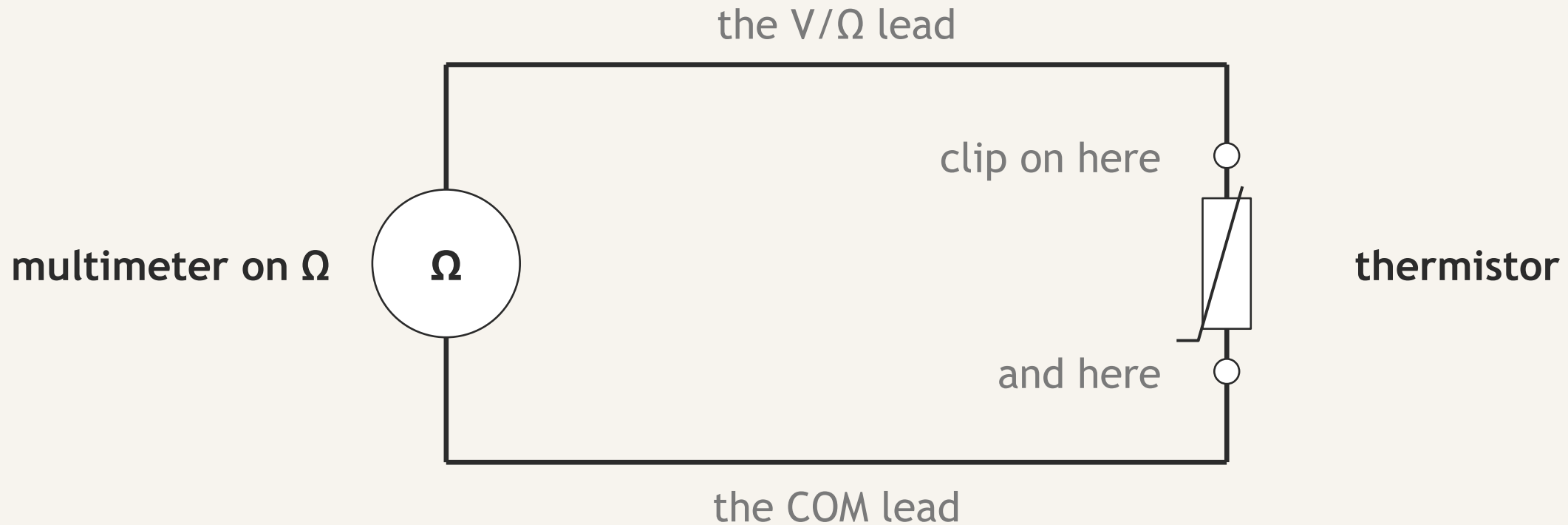
The department has a box of thermistors that lost their labels. Nobody now working here knows what they are. Find out — and write it down so the next person does not have to.

What you get

Item	What it is for
The unknown thermistor	the component under test
A multimeter	measures its resistance
Two leads, 4 mm plugs	meter to component
Two crocodile clips	grip the component's legs

No breadboard — the bead hangs in the clips and goes in the water.

Connecting it up



The clips must not touch — that reads the clips, not the component.

A manual meter does not choose for you

Set the dial to resistance, then pick the range

Range on the dial	It can read up to
200	200 Ω
2k	2 000 Ω
20k	20 000 Ω
200k	200 000 Ω
2M	2 000 000 Ω

Choose, test, adjust

The display says	What it means	What you do
1 or OL	bigger than this range	go up one range
0.02 or 004	far below this range	go down one range
4.72 or 18.6	the range fits	record it, with its unit

Check the range at every temperature — it crosses two or three of them.

Show me — which range?

SHOW ME BOARDS

- 1 You expect about 5 k Ω . Which range do you set?
- 2 On the 2k range the display shows 1. What now?
- 3 On the 200k range the display shows 004. What now?
- 4 You expect about 150 Ω . Which range?
- 5 20k gives 4.72 and 200k gives 004. Which do you record?

Show me – which range?

SHOW ME BOARDS

EXTRA PRACTICE

- 1 The display shows 0.00 on every range. What is wrong?
- 2 You expect about 800 k Ω . Which range?
- 3 Why does 004 on 200k tell you less than 4.15 on 20k?
- 4 The reading drifts upwards as you watch. Why?
- 5 Why must the unit be written down as well as the number?

What you change, and what you keep the same

Change

The temperature of the water the bead sits in.

Measure

The resistance, in ohms, once the reading settles.

Keep the same

The component, the meter, the leads and the clips.

Your turn

Sensor Investigation sheet – Stage 2

Write your method, your equipment list and your risk assessment.
Then swap with another bench and check theirs.

Sensor Investigation sheet, Stage 2

Checkpoint

- 1 Name the two things a manual meter makes you decide that an autoranging one does not.
- 2 Why does the range have to be checked again at every temperature?
- 3 Give one hazard in this practical and what you will do about it.

Daily Review

Part 2 — Planning the test

- 1 Which range would you set for a resistance of about 5 k Ω ?
- 2 The meter shows 1. What does that mean?
- 3 What holds the component in this practical?
- 4 Why must the two clips not touch?
- 5 What must be written down as well as the number?

PART 3

Taking the Readings

Data you cannot trust is worse than no data, because you will believe it.

By the end of this part...

- 1** I can measure resistance safely at a known temperature
- 2** I can record a reading with the correct unit every time
- 3** I can repeat a reading and spot one that disagrees

Before you collect the kettle

SAFETY

Water no hotter than 80 °C. Beaker half full, on a tray, away from the bench edge.

The meter and the leads stay out of the water.

Dry hands before touching the meter. Report every spill.

Three things that cost people their data

The trap	What to do
The range changes under you	check the range at every new temperature
The bead has not caught up	wait until the reading stops drifting
The top of the beaker is hotter	stir, then read meter and thermometer together

Your turn

Sensor Investigation sheet – Stage 3

Six temperatures, from iced water to the hottest safe water.
Every reading twice. Every mean converted to ohms.

Sensor Investigation sheet, Stage 3

What your table should show

ANSWERS

The trend and the size, not a set of numbers to copy

Condition	Resistance
Iced water	the highest reading you take
Room temperature	middling
Held in your hand	lower again
Hottest safe water	the lowest reading you take

Your own numbers are the evidence. Nobody else's will match.

Daily Review

Part 3 — Taking the readings

- 1 What did you do when the display showed 1?
- 2 Why was every reading taken twice?
- 3 Why must every mean be converted to ohms before plotting?
- 4 Does resistance rise or fall as the thermistor warms?
- 5 What is the unit of temperature you recorded?

PART 4

Finding Out What It Is

A measurement becomes a specification only when you can say what part it belongs to.

By the end of this part...

- 1 I can plot my results on a logarithmic graph
- 2 I can compare my curve with the printed ones
- 3 I can find and judge a manufacturer's data sheet

Plotting on the printed graph

Step	What you do
1	Plot each mean resistance as a small neat cross
2	Circle a point that sits off the run – never rub it out
3	Draw one smooth curve, not dot-to-dot
4	Read your answers off the curve, not off a point

Now name it

Read your own curve

State the resistance of your component at 25 °C.

Compare with the six

Say which printed curve yours lies closest to, and how you know.

Look up the part

A measurement plus a data sheet is a specification

- A data sheet is what the maker publishes about the part.
- Search the value you measured: “NTC thermistor 10k data sheet”.
- Check two things: the resistance at 25 °C, and the B value.
- No curve and no table on the page? Keep looking.

Which sources can you trust?

Use	Do not use
The manufacturer's own data sheet	A forum answer with no source
A component distributor's data sheet	An AI chatbot's summary
The exam board's printed graph	A shopping listing with no data sheet

A source that cannot show where its numbers came from is not a source.

Your turn

Sensor Investigation sheet – Stage 4

Plot both thermistors. Name the closest printed type.
Then look the part up and record where you found it.

Sensor Investigation sheet, Stage 4

Checkpoint

- 1 What two numbers does a thermistor data sheet give you first?
- 2 Why is “10 k Ω ” on its own not a specification?
- 3 Give one reason a real component may not sit exactly on a printed curve.

Daily Review

Part 4 — Finding out what it is

- 1 Which source would you trust: a manufacturer or a forum post?
- 2 What does the B value describe?
- 3 Where does a point you rejected go?
- 4 Why is a curve of best fit drawn rather than dot-to-dot?
- 5 State the resistance your own component had at 25 °C.

PART 5

Evaluating and Applying

An engineer says how good the evidence is, not only what it showed.

By the end of this part...

- 1 I can name real sources of error in my own test
- 2 I can choose a component against a stated requirement
- 3 I can justify that choice from my own measurements

Real sources of error in this test

Effect	What it does
Self-heating	the meter's own current warms the bead
Thermal lag	the bead has not reached the water yet
Meter tolerance	the meter is accurate only to a stated %
Reading the scale	a point between gridlines is easy to misplace

Show me — choose the component

SHOW ME BOARDS

- 1 A greenhouse needs about $10\text{ k}\Omega$ at $20\text{ }^{\circ}\text{C}$. Which printed type?
- 2 Is your own thermistor suitable for that? Say why.
- 3 Why can this graph not answer a question about $400\text{ }^{\circ}\text{C}$?
- 4 Two thermistors both read $5\text{ k}\Omega$. What else would you check?
- 5 What would you write on the label going back in the box?

Your turn

Sensor Investigation sheet – Stage 5

Name the three effects that most affected your results.
Then choose a component for the brief, and justify it.

Sensor Investigation sheet, Stage 5

Extension – the LDR

Teacher-led. Not in the pupil sheet.

Thermistor	LDR
Input is temperature	Input is light level
Measured with a thermometer	Measured with a light meter
Measured in °C	Measured in lux
Plotted on the printed graph	Needs a log grid of its own

Both fall as their input rises. Zero lux cannot be plotted.

Plenary

- 1 Why is a resistance axis logarithmic?
- 2 How do you choose a range on a manual multimeter?
- 3 What does an NTC thermistor do as it gets hotter?
- 4 What makes a source reliable?
- 5 What goes on the label of a component you have identified?

- 1 State the four steps for reading a characteristic graph.
- 2 What does the display showing 1 tell you?
- 3 Why is each reading repeated?
- 4 Name two real sources of error in this test.
- 5 Why is a measurement on its own not a specification?

LOOKING AHEAD

Voltage Dividers

Turning that changing resistance into a changing voltage.

What you should be able to do now

- Read a resistance off a logarithmic graph, with its unit.
- Choose, test and adjust the range on a manual multimeter.
- Plan a fair test and write it so someone else could run it.
- Plot your own data and compare it with a published curve.
- Judge whether a source is worth trusting.

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